

An Edge-Deployable YOLOv8n-Based Indian Currency Recognition System with Offline Speech Assistance

Chettiyar Vani Vivekanand

*Computer Science and Engineering
Rajalakshmi Engineering College, Chennai,
India
chettiyarvanivivekanand@rajalakshmi.edu.in*

S Sanjay Krishna

*Computer Science and Engineering
Rajalakshmi Engineering College, Chennai,
India
230701288@rajalakshmi.edu.in*

Dhaanya GK

*Computer Science and Engineering
Rajalakshmi Engineering College,
Chennai, India
230701069@rajalakshmi.edu.in*

Saunak Ramiya Sebasan

*Computer Science and Engineering
Rajalakshmi Engineering College,
Chennai, India
230701298@rajalakshmi.edu.in*

Fareed Ahamed KM

*Computer Science and Engineering
Rajalakshmi Engineering College,
Chennai, India
230701087@rajalakshmi.edu.in*

Abstract—This paper presents a real-time currency denomination detection system based on the YOLOv8n object detector, which is designed to recognize not only different denominations but also old and new design variants of Indian currency notes, resulting in an eleven-class detection scheme. For visually impaired people, independently identifying the denomination of a currency note can be a persistent everyday challenge, especially when older and newer versions of the same denomination are in circulation because differences in size, color, and printed features can affect how reliably a recognition system can identify them. The model was trained on a custom-annotated dataset of 482 images and achieved a precision of 0.98, recall of 1.00, and mAP@0.5 of 0.995 on the validation set. On CPU-only hardware, the system has a total inference latency of about 134 milliseconds per image, indicating its potential for low-cost deployment without dedicated GPU acceleration. An offline text-to-speech engine is used to convert the detected denomination into spoken output, which is especially useful for assistive applications. Experimental testing on unseen images consistently produced high-confidence detections with an average of 0.96. These findings demonstrate that a lightweight model can be used to achieve fine-grained recognition of currency note variants while maintaining real-time performance. As a result, the suggested system offers a workable and repeatable basis for assistive currency-recognition applications that can be implemented on edge devices with limited resources.

Index Terms—assistive technology, currency recognition, YOLOv8, object detection, text-to-speech, visually impaired, edge inference

I. INTRODUCTION

A significant portion of the world's population still suffers from visual impairment, and one of the many daily tasks made more challenging by this condition is identifying paper money. Unlike digital payments, physical banknotes do not offer a direct way for a visually impaired person to independently confirm their denomination without outside assistance. This problem is made more difficult in currencies like the Indian Rupee, where older and newer design variants of the same denomination may continue to circulate at the same time,

differing in size, color scheme, and printed security features. Because they offer a good balance between detection speed and accuracy, YOLO-based object detectors have become particularly useful for this type of application.

Previous research has explored currency denomination recognition extensively using deep learning techniques; systems based on YOLOv5 have demonstrated strong performance for Indian currency recognition, including under cluttered and visually degraded conditions [1], and lighter YOLOv5 Nano variants have demonstrated that real-time inference can be achieved on constrained hardware, making them suitable for edge-based deployment [2]. YOLOv8 and speech synthesis have been combined in more recent work to combine currency detection and audio feedback into a single low-latency pipeline, enabling the communication of denomination and validity information through spoken output [3]. This YOLOv8-plus-speech approach is especially conceptually similar to the system described in this paper, and thus serves as a main point of comparison.

Despite these advancements, the majority of current methods define currency classes based only on face value and do not specifically differentiate between different design generations of the same denomination. This distinction is crucial for currencies that are periodically redesigned because older and newer notes may have distinctly different visual characteristics while still representing the same denomination; as a result, a model trained solely on current-series notes may perform unreliably when presented with older notes that are still in circulation, and the same problem may arise in the opposite direction. Instead of assuming that all notes belonging to the same denomination have the same visual appearance, the suggested approach treats old and new variants of a denomination as distinct classes during both dataset annotation and model training. In order to keep the entire system self-contained and appropriate for assistive use, this paper presents a currency denomination detection system based on YOLOv8n

that follows an eleven-class scheme and explicitly separates old and new variants for applicable denominations. The detection system is combined with a lightweight offline text-to-speech engine that translates the detected denomination into spoken feedback without requiring an internet connection. The model is evaluated on a validation set of currency images, with special attention paid to per-class detection confidence, precision, recall, and inference latency on CPU-only hardware. The purpose of these measurements is to determine whether the suggested system can deliver accurate and fast enough performance for low-cost, real-time assistive deployment.

The remainder of this paper is organized as follows. Section II reviews related work on currency recognition for visually impaired users. Section III describes the dataset, model architecture, and training methodology. Section IV presents the experimental results. Finally, Section V concludes the paper by discussing the limitations of the proposed system and possible directions for future work.

II. RELATED WORKS

Many systems have been proposed to help blind or visually impaired users identify currency notes. Early methods relied on traditional image processing or simple CNN classifiers installed on smartphones or Raspberry Pi devices; more recent work has used modern object detectors.

YOLOv3-based systems achieved real-time banknote detection with reported accuracies around 95–96% on controlled test sets, usually paired with text-to-speech for audio feedback; YOLOv5 and YOLOv8 variants have further improved speed and accuracy on multi-denomination Indian and foreign currency datasets, frequently reaching mAP@0.5 values above 0.95 when trained on well-annotated collections.

A common limitation of these systems is that they concentrate on identifying a single note in isolation and stop at voice announcement of its denomination; relatively little attention has been paid to recognition under viewpoints typical of wearable, hands-free capture or to explicitly modeling the visual differences between old and new note designs for the same denomination, both of which are addressed in this work.

Some works combine currency recognition with obstacle detection or general object recognition on wearable platforms; offline text-to-speech libraries are commonly used to keep such systems operational. In addition to providing a useful wearable capture architecture that most previous recognition-only systems do not provide, this work expands on the robust recognition performance of lightweight YOLO models by explicitly differentiating between old and new note variants during dataset annotation and training.

III. METHODOLOGY

A. System Architecture

The system uses a capture-and-compute pipeline for hands-free operation, as shown in Fig. 1. A camera unit installed on smart glasses records a frame of the currency note and sends it

to a companion smartphone application, which preprocesses the image (resizing and normalizing), passes the frame through a lightweight YOLO-based detector to identify the denomination label, and scores the prediction's confidence. The detected label is then mapped to a corresponding spoken phrase, synthesized using an on-device text-to-speech engine, and streamed to the user via Bluetooth-connected earphones or a speaker. Any reliance on network connectivity is eliminated by keeping the recognition and speech-synthesis phases fully on-device, which is crucial for a wearable assistive technology meant for constant, practical use.

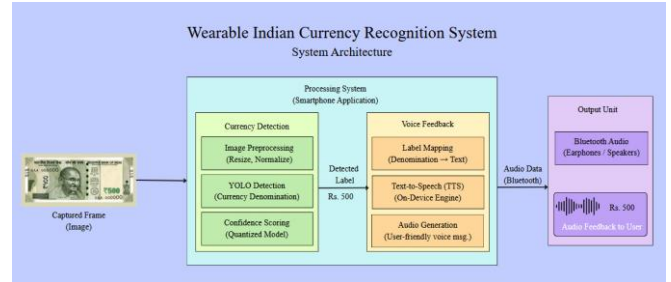


Fig. 1. System architecture: a camera unit on smart glasses captures the note image, which is preprocessed and passed through the YOLO detector; the resulting label is converted to speech and streamed to the user over Bluetooth.

B. Data Collection and Preprocessing

Eleven classes were defined covering both new and old series notes: 10, 20 New, 20 Old, 50 New, 50 Old, 100 New, 100 Old, 200, 500 New, 2000, and the corresponding remaining variant. A representative training batch with bounding-box annotations is shown in Fig. 2, and the per-class instance counts along with the spatial distribution of the annotated boxes are summarized in Fig. 3.



Fig. 2. Representative augmented training batch with ground-truth bounding-box annotations.

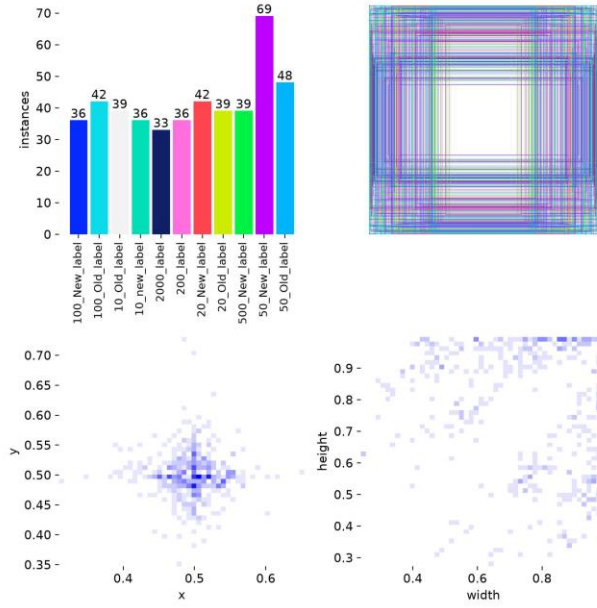


Fig. 3. Dataset statistics: per-class instance counts (top-left), aggregate box positions (top-right), and normalized box center and size distributions (bottom row).

The dataset was augmented for real-world robustness (lighting variation, motion blur, partial occlusion, reflective surfaces and crumpled notes). Identical preprocessing (resizing to 640×640 and normalization) is applied at both training and inference time so that the model sees consistent conditions in both stages.

C. Model Selection and Training

A YOLOv8n (nano) backbone was specifically chosen for deployability rather than peak accuracy alone. Training began with COCO-pretrained weights and ran for 100 epochs on CPU (batch size 16, image size 640). Standard Ultralytics augmentations (mosaic, HSV jitter, horizontal flip, scale, and translate) were used throughout training. Following training, the model was exported to the ONNX format (11.7 MB) for effective on-device inference. At inference time, a confidence threshold is applied, which causes a recapture request rather than informing the user of an uncertain result.

D. Evaluation Protocol

Detection metrics were calculated using a held-out validation set of 23 images; end-to-end latency is measured on the target hardware throughout the entire pipeline, including capture, preprocessing, inference, confidence check, speech synthesis, and audio playback; a separate real-world evaluation pass under live demo conditions (varying lighting, angles, and occlusion) is planned to report practical accuracy distinct from benchmark numbers.

IV. RESULTS AND DISCUSSION

A. Detection Performance

On the 23-image validation set the trained YOLOv8n model achieved:

- Precision: 0.98
- Recall: 1.00
- mAP@0.5: 0.995
- mAP@0.5:0.95: 0.91

All 11 classes reached mAP@0.5 = 0.995, as reflected in the precision-recall curve of Fig. 4. Per-class mAP@0.5:0.95 ranged from 0.846 (100 Old) to 0.995 (50 New).

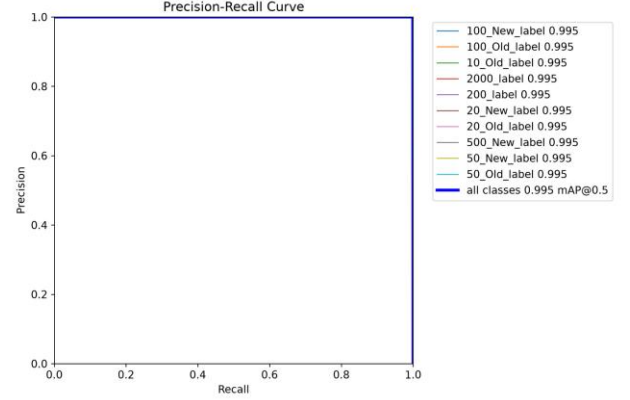


Fig. 4. Precision-Recall curve for all eleven currency classes on the validation set.

Using the exported ONNX model, an independent detection test on the 23 validation images yielded a 100% detection rate (23/23), with minimum confidence of 0.932, maximum confidence of 0.990, and average confidence of 0.960. Fig. 5 displays representative qualitative detections, each annotated with the predicted class label and confidence score.



Fig. 5. Qualitative detection results on validation images, annotated with predicted denomination and confidence score.

B. Training Behaviour

There was no early stopping triggered during the entire 100 epochs of training. As seen in Fig. 6, box loss, classification loss, and DFL loss all steadily declined throughout training, but precision, recall, mAP@0.5, and mAP@0.5:0.95 all improved and stabilized in the later epochs. By epoch 11, mAP@0.5 reached the 0.995 ceiling and stayed steady, but mAP@0.5:0.95 kept improving to 0.91, suggesting that bounding-box precision was being refined rather than overfitting. The risk of memorization is further decreased by the model's small capacity (roughly 3M parameters) in comparison to the 459 training images.

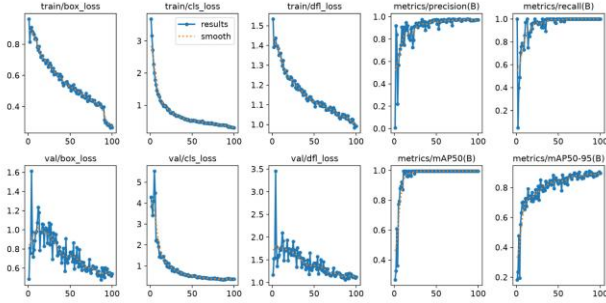


Fig. 6. Training and validation loss curves (box, classification, DFL) together with precision, recall, mAP@0.5 and mAP@0.5:0.95 over 100 training epochs.

Fig. 7 shows the confusion matrix on the validation set, normalized by true class. Confusion between different denominations and between old and new variations of the same denomination is minimal, as evidenced by the fact that all eleven classes are nearly entirely on the diagonal.

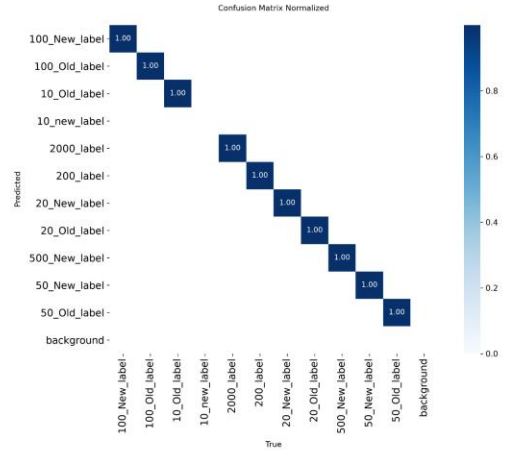


Fig. 7. Normalized confusion matrix on the validation set.

C. Inference Latency

Preprocessing took 1.7 ms, inference took 131 ms, and post-processing took 1.4 ms on the training CPU, for a total of roughly 134 ms per image. This number should significantly decrease on a smartphone with GPU or NPU acceleration, enabling real-time operation at several frames per second.

D. Voice Feedback

For each of the eleven denominations, the offline text-to-speech engine produced speech with success. The hands-free, low-latency design goal is achieved with audio generation latency of about 0.5 seconds per phrase and no need for an internet connection.

E. Discussion

The findings show that a lightweight YOLOv8n model can achieve detection performance comparable to the literature while still being appropriate for mobile inference when trained on a carefully prepared dataset that explicitly distinguishes between old and new note variants. The high validation metrics and 100% detection rate on the test images attest to the successful recognition of Indian currency notes under the assessed conditions, including both their old and new design variants.

The current recognition pipeline does not address the open direction of extending the system to track cumulative transaction amounts, confirm that a physical note has left the capture frame, or otherwise confirm that a cash exchange has been fully completed.

The small validation set size, the absence of a fully held-out test split, and the still-limited evaluation under extreme real-world conditions (heavy occlusion, severe motion blur, worn notes) are limitations that need to be addressed in future work. Recognized gaps serve as the foundation for the upcoming experimental campaign.

F. Comparison with Existing Studies

In order to contextualize these findings, Table I contrasts the suggested system with representative currency-recognition studies published by IEEE, Springer, and other venues [1]–[8].

The comparison is indicative rather than a like-for-like benchmark because these studies differ in currency, dataset, model, and evaluation metric.

TABLE I
COMPARISON WITH EXISTING CURRENCY RECOGNITION SYSTEMS

Ref.	Method	Currency and data	Reported result
[1]	YOLOv8 with gTTS speech	Counterfeit Indian notes	Fake-note check with voice output
[2]	YOLOv3	INR, 3,720 images, 7 classes	Detection 95.71%, recognition 100%
[3]	YOLOv8n with SE blocks	USD, EUR, BDT, 30 classes	mAP@0.5 97.21%, 3.1 ms per image
[4]	YOLOv9 on smart glasses	PKR, 3,611 images, 7 denominations	Accuracy up to 95%
[5]	CNN compared with YOLOv5	INR	CNN above 79.83% accuracy, about 1% below YOLOv5
[6]	YOLOv5	INR, 10,000 images	Average accuracy 92.71%
[7]	IPCRNet (CNN classifier, 3.6M parameters)	INR, 50,263 images, 11 classes	Average accuracy 96.75%
[8]	Three-stage Faster R-CNN (VGG16)	KRW and JOD notes and coins	Multi-stage detection of notes and coins
This work	YOLOv8n (11.7 MB ONNX)	INR, 482 images, 11 classes	P 0.98, R 1.00, mAP@0.5 0.995, 134 ms on CPU

Table I makes three observations: First, the detection quality of the suggested model (precision 0.98, recall 1.00, mAP@0.5 of 0.995) is on par with or better than the results reported for the YOLO-based systems in [2], [3], and [6], which report a detection accuracy of 95.71%, an mAP@0.5 of 97.21%, and an average accuracy of 92.71%, respectively, and it also compares favorably with the 96.75% average accuracy of the lightweight classifier in [5] is also consistent with the use of a single-stage YOLO detector in place of the multi-stage pipeline of [8].

Second, the proposed system divides old and new note designs into eleven classes, while the YOLO-based Indian-currency systems [2] and [6] are based around seven denominations. Of the works compared here, [7] also models old and new notes as separate classes, but it is an image classifier and does not localize the note.

Third, the system is completely offline and lightweight. The YOLOv8n model has about 3M parameters and an 11.7 MB ONNX export, which is similar in size to the 3.6M-parameter network of [7] while also localizing the note. Speech is generated by an on-device engine, while [1] uses the gTTS library, which calls an online text-to-speech service. The end-to-end latency of about 134 ms per image reported in [3], but that difference may be due to hardware, and it is still sufficient for real-time use, as covered in Section IV-C.

The size of the current evaluation must be taken into consideration when reading these comparisons; for example, [7] is tested on approximately 9,500 images, while the suggested figures are obtained on a validation set of 23 images (Section IV-A). Therefore, a larger held-out test set is necessary before the aforementioned advantages can be deemed conclusive, as mentioned in Section IV-E.

V. CONCLUSION

In this paper, a real-time Indian currency denomination detection system for visually impaired users was presented. Precision 0.98, recall 1.00, mAP@0.5 0.995, and mAP@0.5:0.95 0.91 were attained by a YOLOv8n model trained on an 11-class dataset that clearly separates old and new note variants. The model was incorporated into a useful wearable design that includes a camera mounted on glasses and a companion smartphone that uses Bluetooth to provide offline text-to-speech feedback and inference.

The work demonstrates that precise, hands-free note recognition is possible on hardware with limited resources without compromising real-time performance. In order to improve the system's usefulness for separate financial transactions, future research will investigate expanding it to track cumulative transaction amounts and confirm full note exchange.

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